

Exercise 6: Energy International

Goals:

The cadets use the six-step problem solving process to solve a problem

Cadets then discuss their experiences in using the techniques and working as a group

Group Size:

5 cadets per team

Time:

Approx. 1.5 hours

Material:

- Outline Sheet for the Instructor (with solution)
- Energy International Briefing Sheet
- Energy International Data Sheet #1 – 5
- Candidate Summary Sheet

Process:

1. Introduce the lesson using the outline sheet and divide the cadets as desired into teams of five members each
2. Provide each team with a copy of the exercise (the Briefing Sheet and Candidate Summary Sheet for the team and the Energy International Data Sheet [different one to each of the five cadets])
3. Announce the time allotted to do the problem and signal work to begin
 - o *NOTE. The exercise contains all the necessary information to work the problem so that there is no need to overview the problem in advance*

Energy International Outline Sheet

OPEN ENDED QUESTION: What are the six steps in the problem-solving process?

WRITE ON THE BOARD:

1. Recognize the problem
2. Gather data relative to the problem
3. List possible solutions to the problem
4. Test possible solutions to the problem
5. Select the best solution to the problem
6. Implement the solution

Your ability to solve problems is essential, and problems are solved best when a method or process is used. Studies consistently show that people who approach problems systematically are typically the most successful. You too can have that success – you'll get the opportunity to solve problems using this process throughout your time in ROTC and in the future as an AF officer

Continue with the guided discussion.

SOLUTION:

The key to solving this problem lies in the quality of the communication, which occurs within each group, as members analyze the information provided. Although the data sheets appear, at first glance, to be identical, small variances do exist, which provide information critical to the solution. There is only one candidate who meets all of the criteria outlined in the data sheet, as the following discussion explains:

The New Mexico Institute of Earth Sciences and St. Francis University require three special subjects for graduation and are therefore smaller than the Massachusetts Institute of Sciences, or the New York School of Mines. St. Francis is not the smallest, therefore New Mexico Institute of Earth Sciences must be. This makes N.M.I.E.S. a women's university. Brazilians hold a feudal attitude toward women.

Seismology and paleontology are essential for General Membership. St. Francis does not offer seismology; therefore, no graduate of St. Francis can qualify for general manager. None of the Brazilian staff understands English, nor do the gov't inspectors; therefore, before the General Manager can countersign the inspectors report, he must be able to read Portuguese.

Each candidate except **Gadolin** is disqualified because he/she lacks the qualifications outline.

Guided Discussion (about 15 minutes)

LOQ: Which of the problem-solving steps were used in solving this problem?	ANTICIPATED RESPONSE: 1. Recognize the problem 2. Gather data 3. List possible solutions 4. Test possible solutions 5. Select the best solution 6. Implement
FUQ: How effective was your method?	Solicit
FUQ: Who emerged as the most effective member of your group?	Solicit
FUQ: Describe the role played by the third person	ANTICIPATED RESPONSE: 1. Dominant/directive 2. Informal group leader 3. Worker or leader, or a combination of both
FUQ: When did your group first realize that the data sheets were different?	How long: immediate, never discovered, or about halfway
FUQ: How effectively did your group communicate, organize, and classify the data	ANTICIPATED RESPONSE: Assess each group on its individual merits 1. Did they use one-way or two-way communication 2. Was the group open or closed to solving the problem? 3. Did they organize quickly? 4. Did they sort the data by topic, school, person, etc.?
FUQ: How might the skills you used in the exercise make you a more effective manager?	This exercise shows that a systematic method to problem-solving; a skill needed is by managers
FUQ: Name some situations in the military environment where you might use the problem-solving process?	Solicit responses and then use your personal experience to give examples

ENERGY INTERNATIONAL BRIEFING SHEET

Instructions to the Group:

- You are a committee made up of the General Managers of Energy International (E. I.)
- You have just flown into town.
- This is the first meeting of the group
- You have just learned that E. I. will open a new Brazilian plant, and your first job is to select a General Manager from among the seven applicants.
- Basically, the information you bring with you is in your head, do NOT show your E. I. data sheet to anyone else

Assumptions which need to be made explicit:

- Assume that there is one solution
- Assume that all data are correct
- You have ½ hour to work the exercise
- Assume that today's date is April 1, 19992
- There must be substantial agreement when the problem has been solved
- You must work the problem as a team

ENERGY INTERNATIONAL CANDIDATE SUMMARY SHEET

NAME: R. Illin
 DATE OF BIRTH: March 2, 1957
 PASSPORT: L3452 – USA
 EDUCATION: NY School of Mines – degree in mineralogy – 1977
 EMPLOYMENT: Research – NY School of Mines – 1878-1986 Lectures –
 Minerology – Univ. of Bonn – 1986-1990 Manager – Utah Cooper
 Mining Co. Plant – 1990 to date
 LANG COMMAND: English, French, German, Portuguese

NAME: S. Hule
 DATE OF BIRTH: May 4, 1949
 PASSPORT: H4567 – USA
 EDUCATION: New Mexico Inst. of Earth Sciences – degree in mineralogy – 1975
 EMPLOYMENT: Uranium Unlimited – management trainee – 1975-1977 Anaconda
 Copper Co., Montana area-geology officer – 1978-1985
 LANG COMMAND: English, French, Portuguese

NAME: T. Gadolin
 DATE OF BIRTH: June 5, 1950
 PASSPORT: L7239 – USA
 EDUCATION: NY School of Mines – degree in mineralogy – 1975
 EMPLOYMENT: United Kingdom Mining b card – management trainee 1975-1977
 Assistant Manager – NDA Cheshire plant – 1978-1986
 Manager – Idaho Cobalt Mining Co. – 1986 to date
 LANG COMMAND: English, Portuguese

NAME: U. Samar
 DATE OF BIRTH: April 6, 1958
 PASSPORT: H6259 – USA
 EDUCATION: Mass Institute of Sciences – degree of mineralogy – 1979
 EMPLOYMENT: Jr. Engineer – W. Virginia Mining Research Station – 1979-1988
 General Manager – Loberian State Mining Plant – 1988 to date
 LANG COMMAND: English, German, Swahili, Portuguese

NAME: V. Lute
 DATE OF BIRTH: April 16, 1955
 PASSPORT: K62371 – USA
 EDUCATION: NY School of Miners – degree in mineralogy – 1976
 Jr. Development Mineralogist – Ontario Mining Constr. Ltd. –
 1976-1970
 Asst. Chief Mineralogy Officer – Canadian Dev. Board – 1980-1983
 Plant Manager – Welsh Mining Co., Ltd. – 1984 to date
 LANG COMMAND: English, French, Welsh, Pekingese

NAME: W. Neddy
 DATE OF BIRTH: H63241 – USA
 PASSPORT: St. Francis University – degree in mineralogy – 1973
 EMPLOYMENT: Asst. Manager – Societe Debunquant D’Algeria – 1973-1977
 Manager – Kemekatka Mining Co. – 1978 to present
 LANG COMMAND: English, Portuguese, Russian, Arabic

NAME: X. Lanta
 DATE OF BIRTH: September 8, 1955
 PASSPORT: Q123YB – Canada
 EDUCATION: University of Quebec – Diploma in English – 1975 Mass.
 Institute of Sciences – degree in mineralogy – 1978
 EMPLOYMENT: Tech Officer, Sardinia Mining Corp. – 1980-1988 Manager – Maob
 Valley Mining Plant – 1988 to date
 LANG COMMAND: Spanish, English, Portuguese

ENERGY INTERNATIONAL DATA SHEET #1

Your group is a committee made up of the General Managers of Energy International, a young, medium-sized, growing organization. The mission of E.I is to locate and develop mineral claims (copper, uranium, cobalt, etc.) ...

The company's business has grown very rapidly especially in South America, where your organization has been Made welcome by the governments. In a recent meeting, the board of directors decided to develop a new property near Fortaleza, in northeastern Brazil. This operation will include both mining and milling production.

The date is April 1, 1992. You have come from your respective plants in different locations. This is the initial session of your annual meeting. Your first order of business today is to select a new General Manager for the Brazilian plant from among the candidates on the attached list.

Fortaleza, Brazil, has a hot climate, one railroad, a scheduled airline, a favorable balance of trade, a feudal attitude toward women, considerable unemployment, a low educational level, a low literacy rate, and a strongly nationalistic regime.

The government has ruled that the company must employ Brazilians in all posts except that of Manager. It has also installed an official inspector, who will take a monthly report which must be countersigned by the company's representative. None of the company's employees or staff can read or write any language but Portuguese.

ENERGY INTERNATIONAL DATA SHEET #2

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The government has ruled that the company must employ Brazilians in all posts except that of Manager. It has also installed an official inspector, who will take a monthly report which must be countersigned by the company's representative, who must be an American citizen.

Fellowship in the Inst. of Mineralogy can be obtained by men over 35 years of age who have otherwise qualified for General Membership in the Inst. St. Francis University, which is not the smallest school, requires the following special courses for graduation: paleontology, geophysics, and oceanography.

ENERGY INTERNATIONAL DATA SHEET #3

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The company's business has grown very rapidly especially in South America, where your organization has been made welcome by the governments. In a recent meeting, the board of directors decided to develop a new property near Fortaleza, in north-eastern Brazil. This operation will include both mining and milling production.

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The government has ruled that the company must employ Brazilians in all posts except that of Manager. It has also installed an official inspector, who will take a monthly report which must be countersigned by the company's General Manager. By law, the General Manager must have had at least three years experience as a manager in charge of a mining operation.

There are a number of schools offering degrees in mineralogy, a degree is essential to qualify for General Membership in the Inst. of Mineralogy. The smaller universities require three, the larger four, of the following special subjects as a part of their graduation requirements: geology, geophysics, oceanography, paleontology, seismology. The smallest is a women's university.

ENERGY INTERNATIONAL DATA SHEET #4

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The government has ruled that the company must employ Brazilians in all posts except that of General Manager. The government has also installed an official inspector, who will make a monthly reports to the government This report must be signed by the company's representative, who must be a Fellow of the Inst. of Mineralogy.

There are a number of schools offering degrees in mineralogy, the most recently founded is the New Mexico Inst. of Earth Sciences. This Inst. was established under a special grant and opened in 1965.

In order to earn a bachelor's degree in mineralogy, this school requires geology, seismology, and paleontology in addition to the usual courses.

ENERGY INTERNATIONAL DATA SHEET #5

Your group is a committee made up of the General Managers of Energy International, a young, medium-sized, growing organization. The prime mission of E.I is to locate and develop mineral claims (copper, uranium, cobalt, etc.)

The company's business has grown very rapidly especially in South America, where your organization has been made welcome by the governments. In a recent meeting, the board of directors decided to develop a new property near Fortaleza, in northeastern Brazil. This operation will include both mining and milling production.

The date is April 1, 1992. You have come from your respective plants in different locations. This is the initial session of your annual meeting. Your first order of business today is to select a new General Manager for the Brazilian plant from among the candidates on the attached list.

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The government has ruled that the company must employ Brazilians in all posts except that of manager. It has also installed an official inspector, who will make a monthly report which must be countersigned by the company's representative. None of the government inspectors can read or write any language but his own.

There are a number of schools offering degrees in mineralogy, but a passing grade in paleontology is essential to qualify for General Membership in the Inst. of Mineralogy. The largest university is the New York School of Mines, which requires the following special subjects for graduations: geology, paleontology, geophysics, and seismology.